

Lecture - 01 (sm)

Date: 26/02/2023

CCE 321: Computer Peripherals and Interfacing

Chapter 01

Peripheral devices: up - v আর্থ externally যাক device connected থাকে।

3 Types:

1. Input Peripheral
2. Output Peripheral
3. Memory Peripheral (RAM, ROM)

Interfacing: external device যুক্ত যাক CPU এর আর্থ connect করতে পারে।

Peripheral characteristics:

- i) Different speed (slow devices CPU এর slow করে)
- ii) Different code, control signals
- iii) Transmit data in serial & parallel form
- iv) Even work at higher voltages than the CPU
- v) All operate at diff slower speeds than the CPU

Peripheral Functions:

- i) Buffering (RAM এর portion নিয়ে কাজ করে)

Spooler - hard disk use করে; printer এর background printing এর জন্য use হয়

ii) Converting data to and from serial and parallel forms. (serial best for short distance & parallel for long distance)

iii) Converting analog, digital form

iv) voltage conversion

v) protocol conversion

vi) handling of status signals

Types of Computer:

1. Mainframe

2. Super Computer

3. Mini Computer

4. Micro Computer

MP data type:

- bit, byte, word

- Unsigned and signed binary integers

- BCD

- ASCII

- Floating Point Numbers

Floating Point Number

- The Conversion Procedure

- 3 parts:
1. sign
 2. exponent
 3. mantissa/significand

* Convert -4.75 to 8-bit floating point format.

$$\Rightarrow (4)_{10} = (100)_2$$

$$0.75 \times 2 = 1.50 \quad 1$$

$$0.50 \times 2 = 1.00 \quad 1$$

$$\therefore (4.75)_{10} = (100.11)_2$$

Normalize:

$$100.11 \times 2^0$$

$$10.011 \times 2^1$$

$$1.0011 \times 2^2$$

$$\text{Mantissa} = 0011$$

$$\text{exponent} = 2 + 3 = 5 = 101_2$$

$$\text{sign} = 1$$

$$(-4.75)_{10} = 1 \ 101 \ 0011 = (D3)_{16}$$

bias k - exponent

$$2^{(k-1)} - 1 = 2^{3-1} - 1 = 3$$

8 bit $k=3$

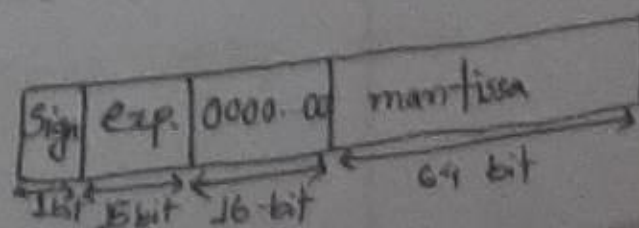
16 " " $k=5$

32 " " $k=8$

64 " " $k=11$

96 " " $k=15$

96 bit $k=15$



* Microcomputer structure & operation

Major parts:

- CPU $\rightarrow$ Registers
- Memory
- I/O
- Buses

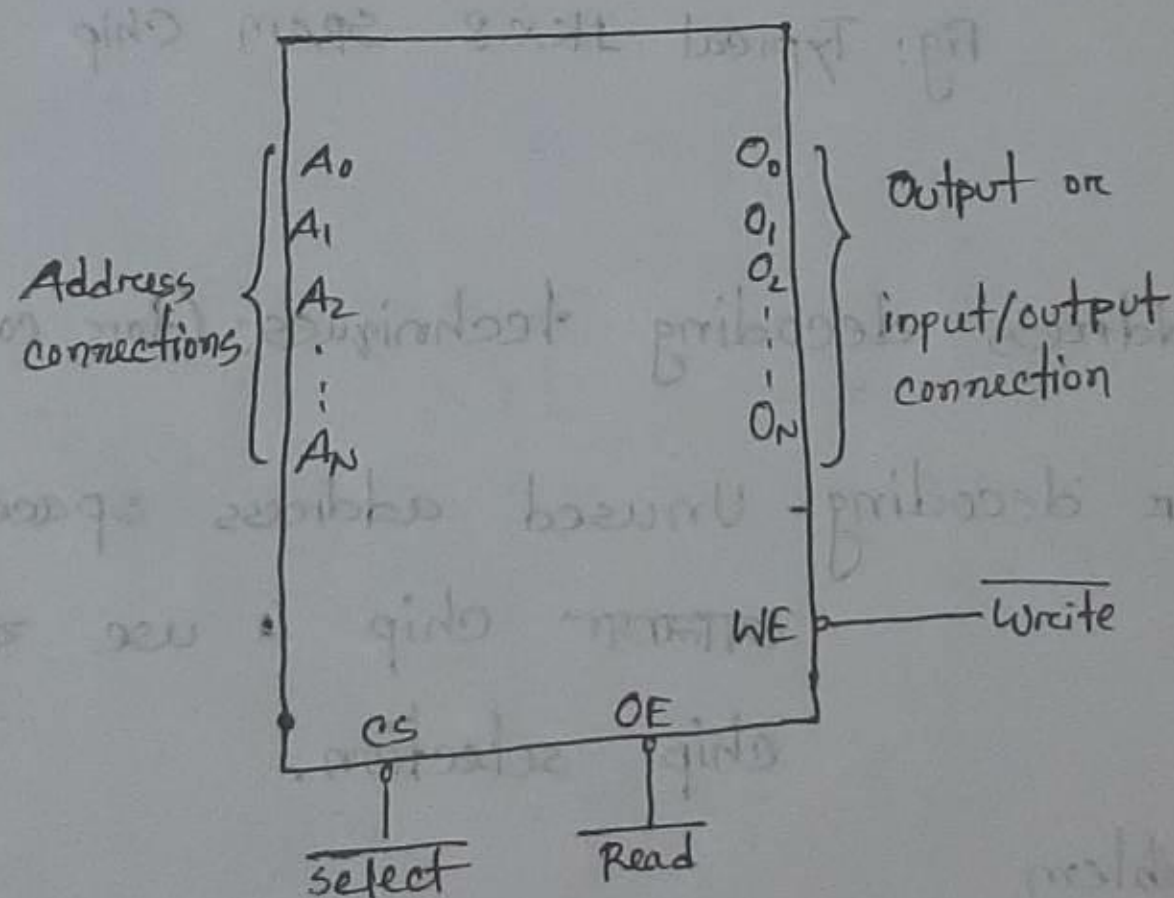
Memory System

Two types memory - RAM, ROM

4 Type Pin Connection:

- 1) Address - passes address
- 2) Data - passes data
- 3) Chip selection - selects which chip to use
- 4) Control (RD/WR)

Memory Pin Connection



Address / Memory Decoding - chip selection এর value দেওয়া-
কোন chip বন্ধন কাজ করবে

Array design - অনেকগুলো memory chip একত্রে interconnect
করা

Main memory Array Design

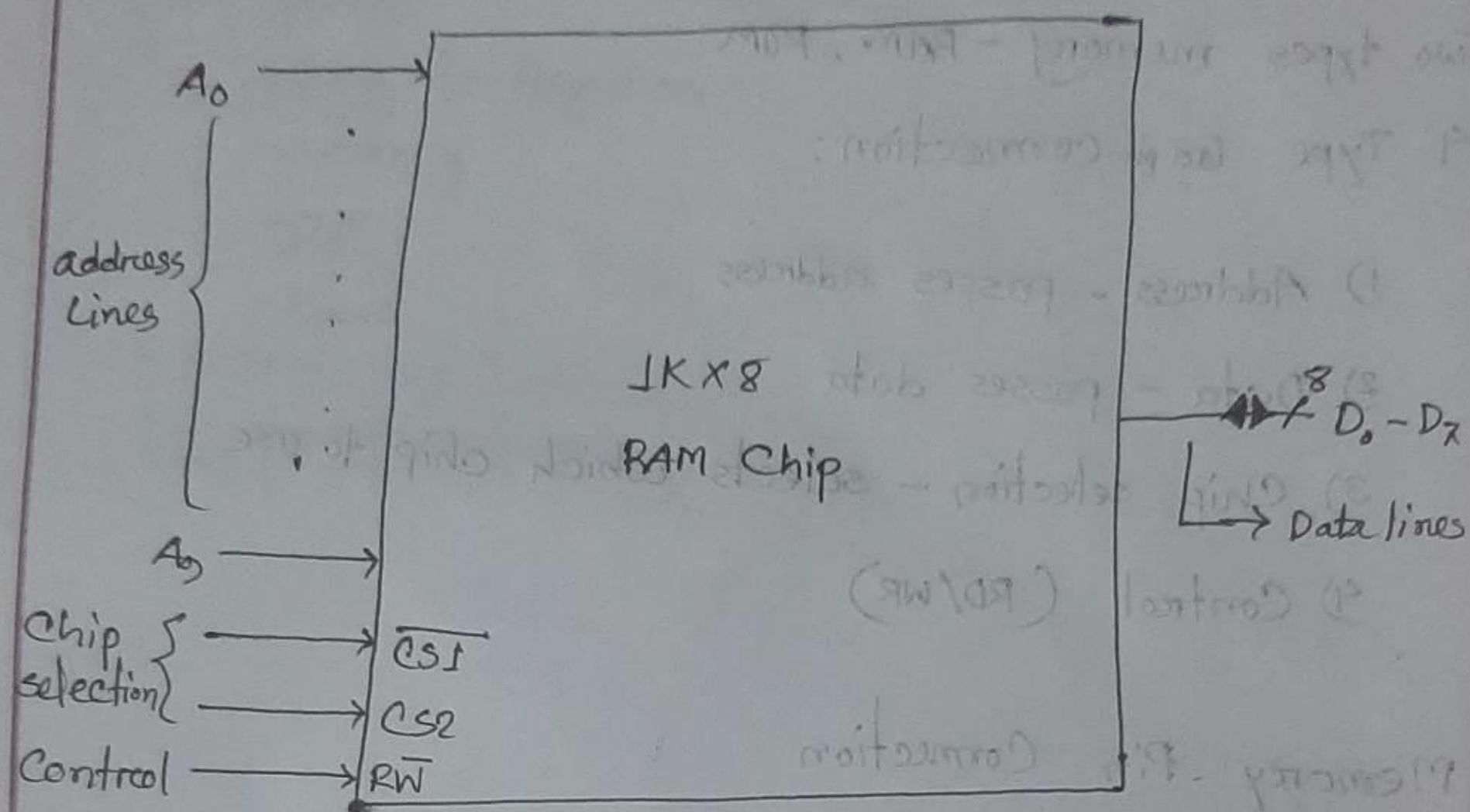


Fig: Typical 1Kx8 SRAM Chip

Three Address decoding techniques: (for connecting up to RAM/RAM)

1. Linear decoding - Unused address space જુલાલે બોલાદા chip & use કર્યા રહ્યાં for chip selection.

Problem

i) આ address utilize રહ્યાં ના (memory capacity વાજા)

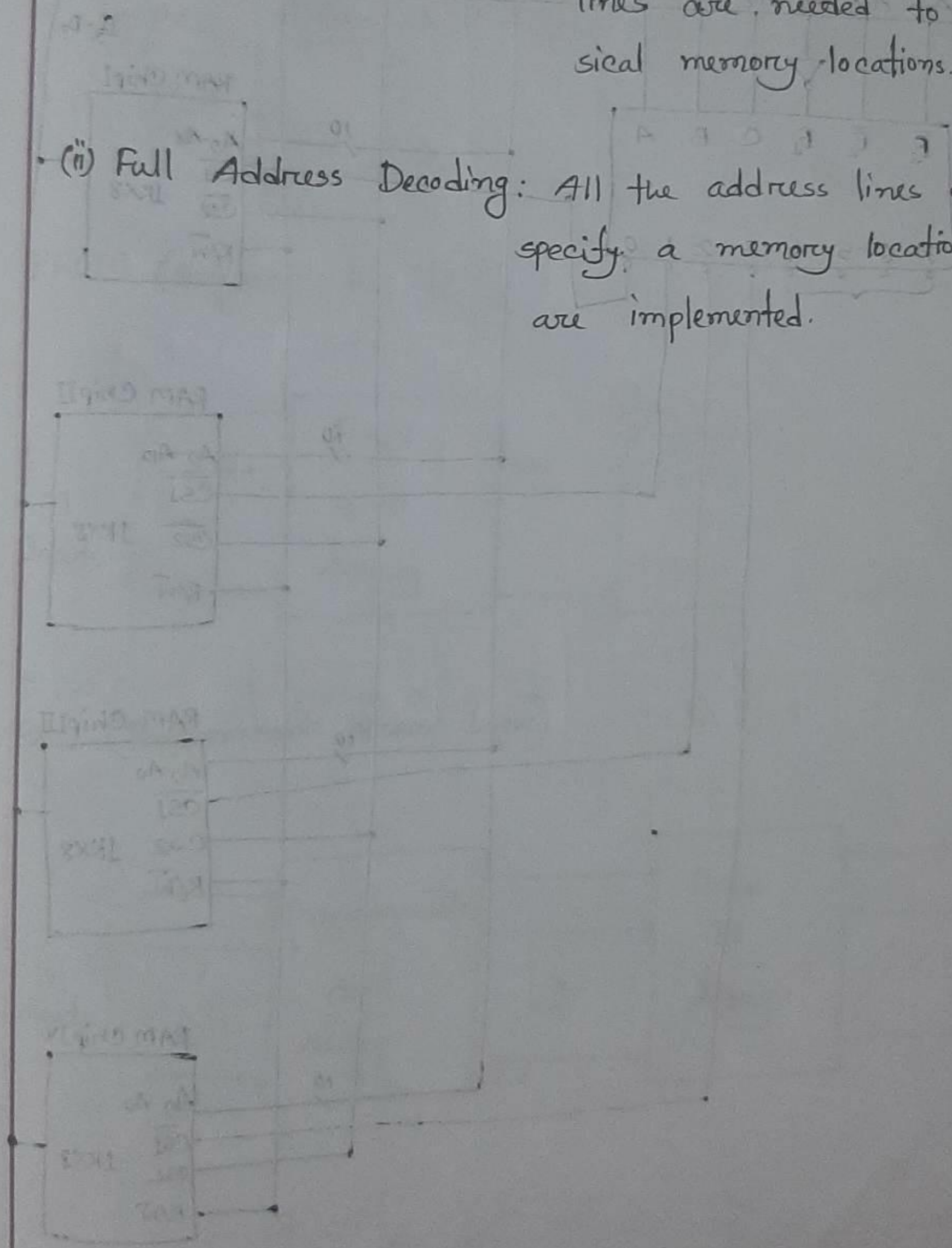
ii) Bus conflict રહ્યાં (અકચારા ૨ ને ૧ ચાલે ના જેમ program વગર દેશ) → solution

iii) seriality maintain વાજા યાસ ના

iv) smaller device & use

2. (i) Partial Address Decoding: not all address space is implemented; a subset of the address lines are needed to point the physical memory locations.

(ii) Full Address Decoding: All the address lines are used to specify a memory location; all addresses are implemented.



Full Decoding

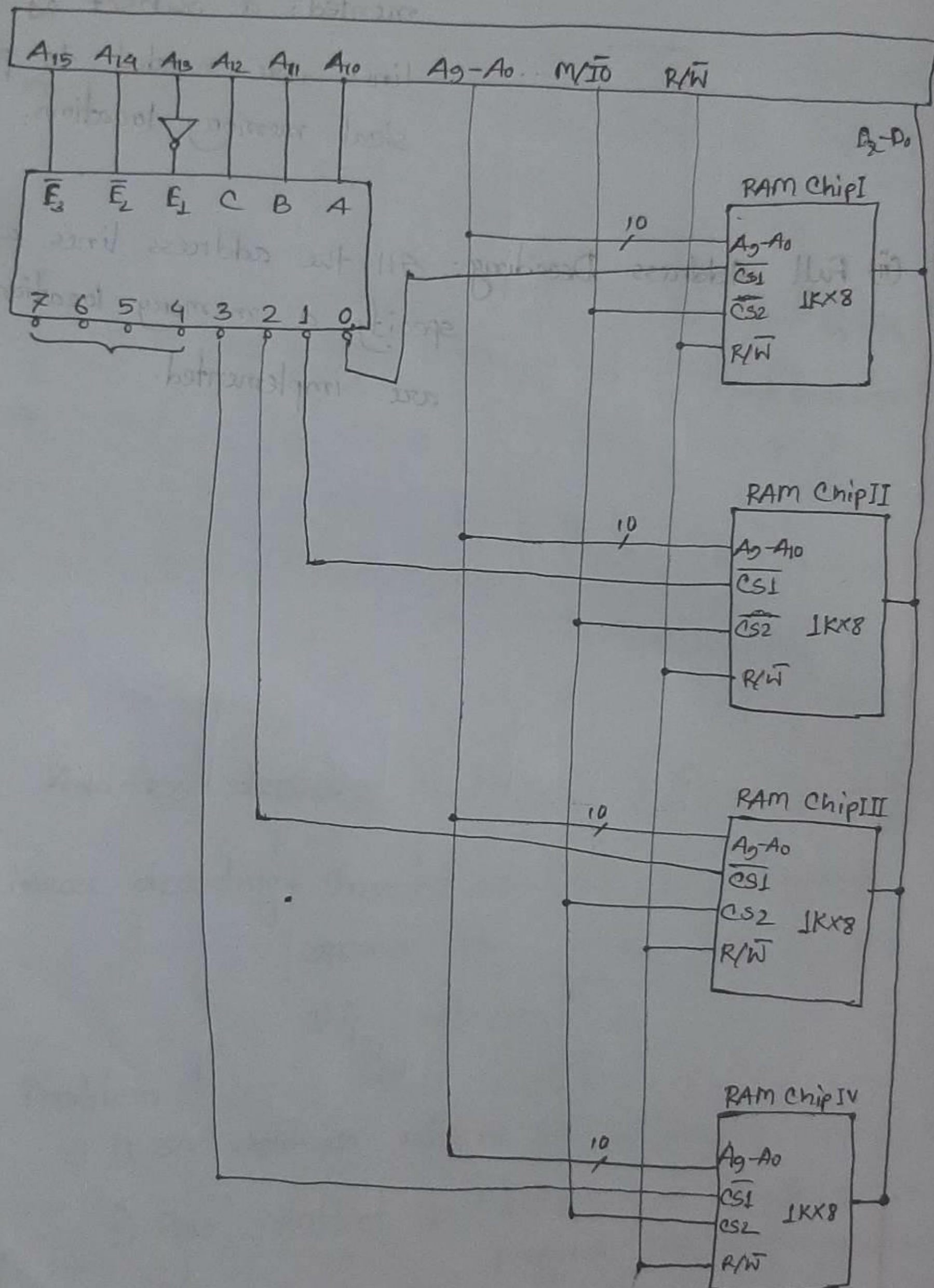
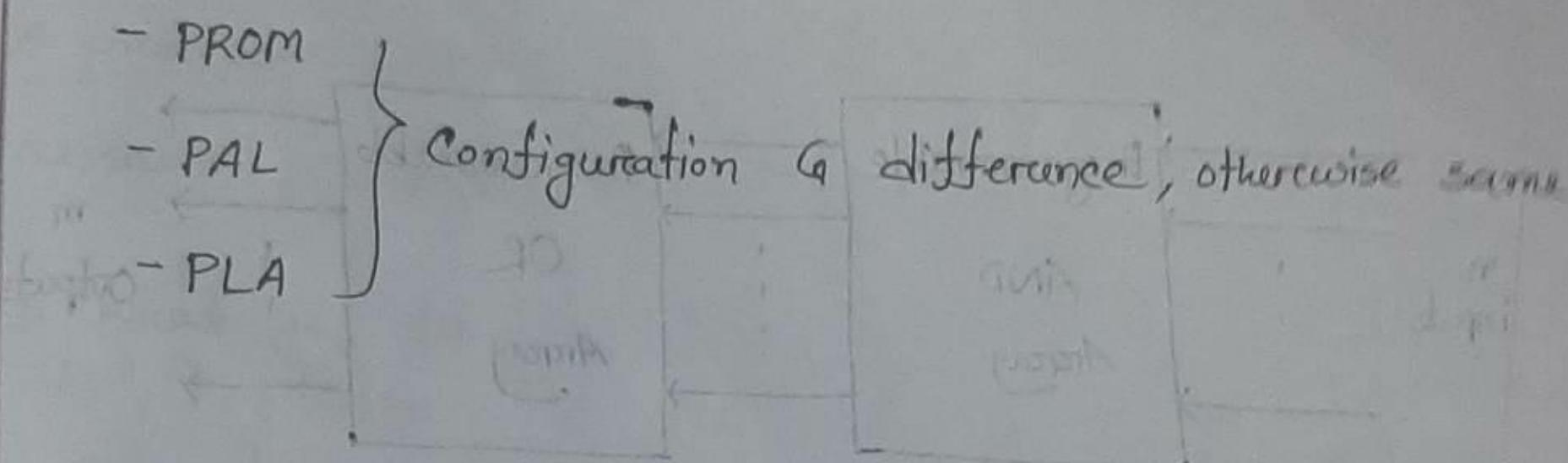
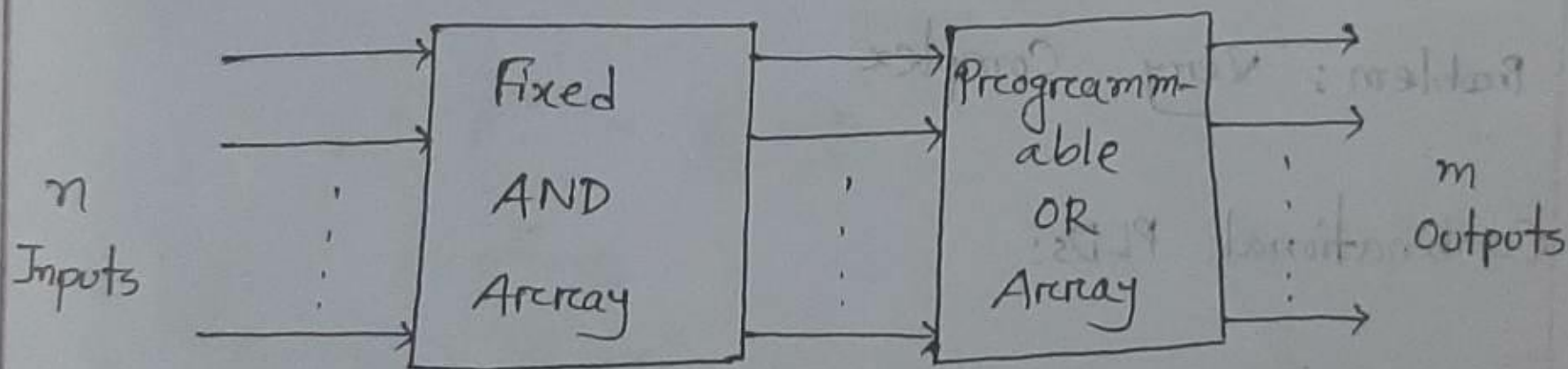


Fig: Interconnecting a μP with a 4K RAM using full decoded memory addressing.

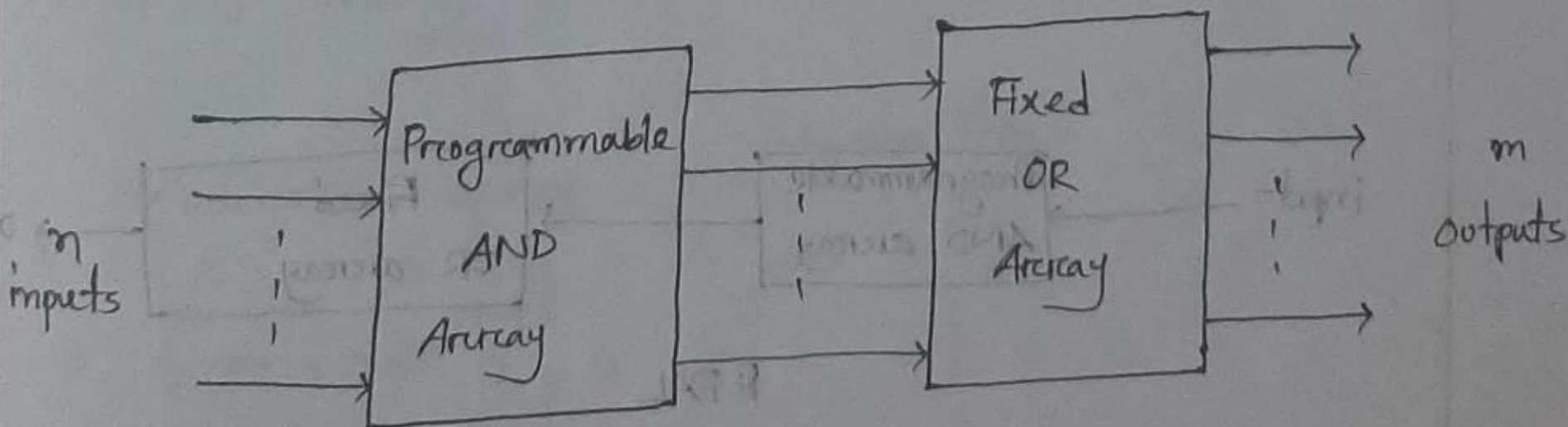
3. PLD (Programmable Logic Devices)



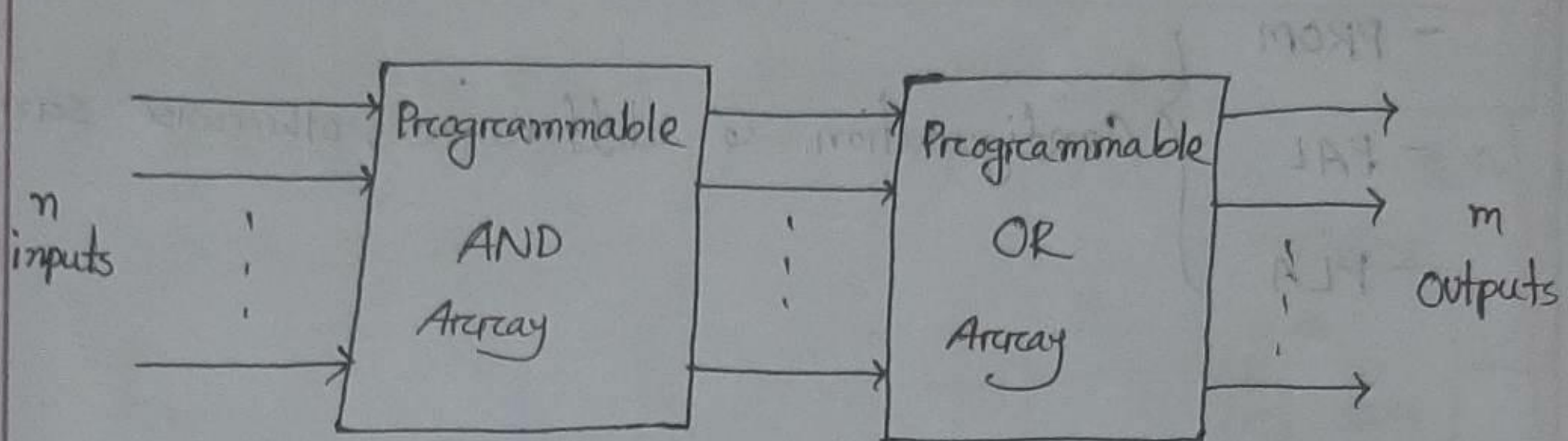
PROM (Programmable ROM)



PAL (Programmable Array Logic)

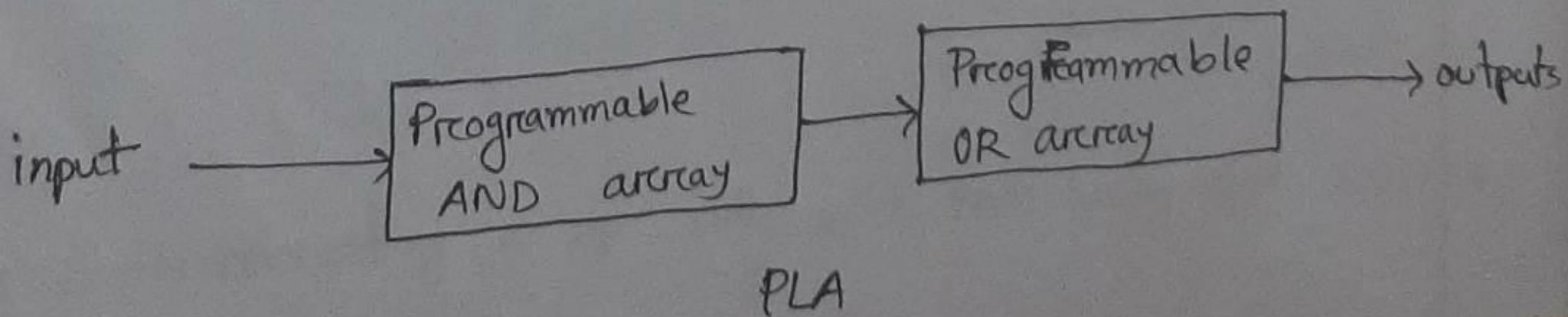
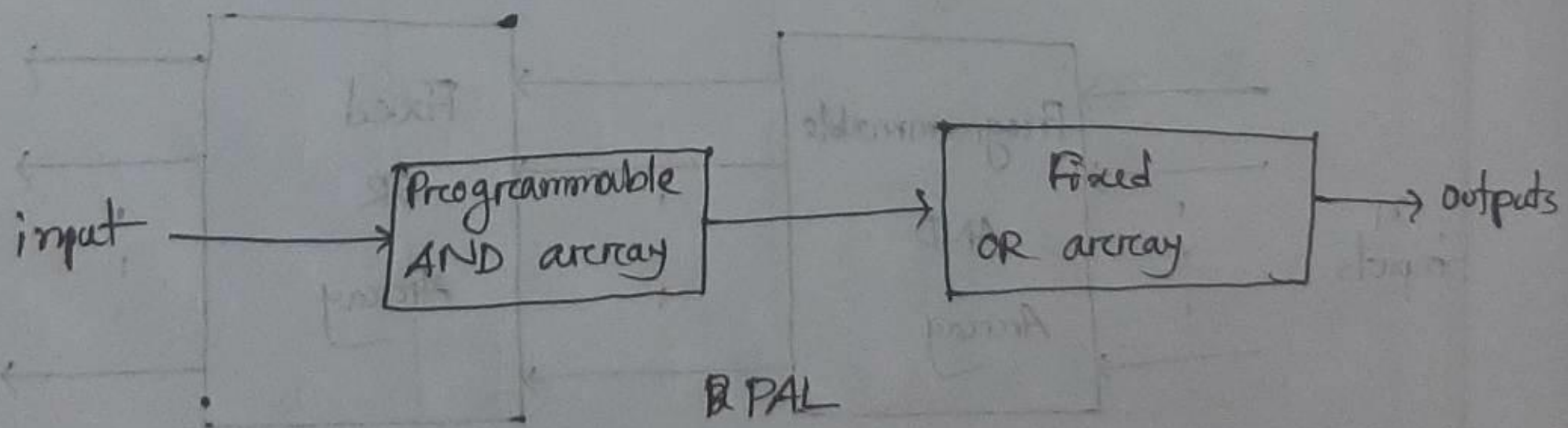
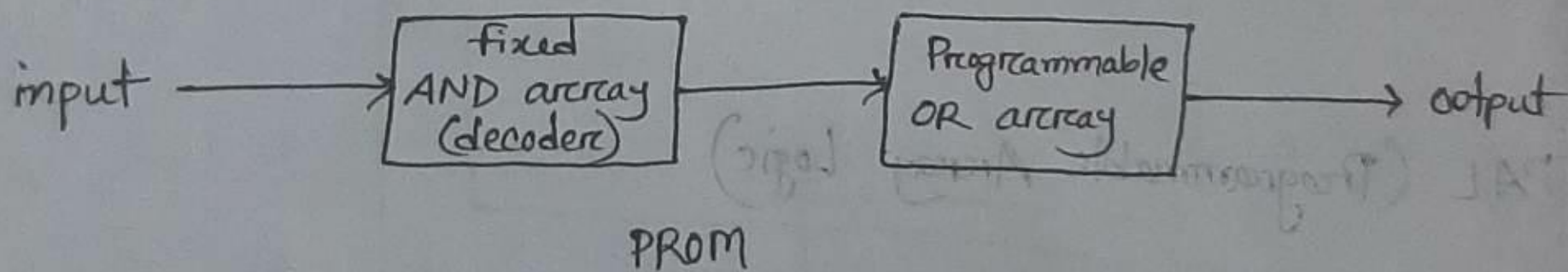


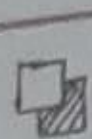
PLA (Programmable Logic Array)



Problem: Very Complex

Combinational PLDs:





Example 01:

A processor having 16 bits address line and it is byte addressable. Interface 4096x8 RAM with the processor

1. What is the memory capacity of the processor?

$$\text{Capacity} = \text{Address Space} * \text{Word size}$$

$$= 2^{\text{Address lines}} * \text{Word size}$$

$$= 2^{16} * 8 \text{ bits}$$

$$= 2^{16} \text{ B}$$

$$= 64 \text{ KB}$$

2. How many address lines are required for RAM?

$$\text{Address space} = 2^{\text{Address lines}}$$

$$\Rightarrow 4096 = 2^{\text{A.L}}$$

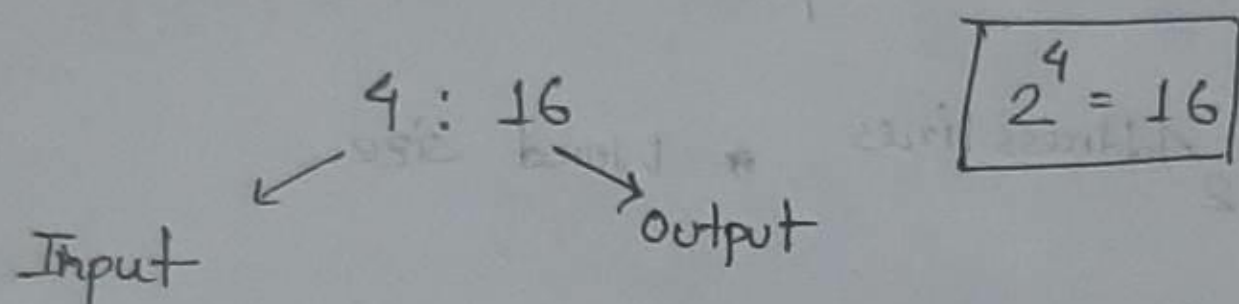
$$\Rightarrow 2^{12} = 2^{\text{A.L}}$$

$$\therefore \text{Address lines} = 12$$

2. $A_{15} = 1, A_{14}$
 value char
 2. A_{12}, A_{13}
 value char
 2. A_{10}, A_{11}
 value char
 2. A_8, A_9
 value char
 2. A_6, A_7
 value char
 2. A_4, A_5
 value char
 2. A_2, A_3
 value char
 2. A_0, A_1
 value char

$$\frac{2^{16} \times 8}{2^{12} \times 8} = 2^4 = 16$$

Don't care
address 4
address-2



$$\begin{array}{l} \text{Base} \quad \quad \quad 4KB \\ \nearrow \quad \quad \quad \uparrow \\ \text{End address} = 8000H + \left(\frac{2^{10} \times 4 - 1}{\text{Dec, mtd}} \right) \\ = 8000H + (1023) \\ = 8000H + 03FFH \\ = 83FFH \quad 0FFFH \end{array}$$

$$4K \times 8 = 2^{12}$$

[illegible]

RAMS 170 ←
chip selection
यूसा चाल यदि
A15 0 1 23

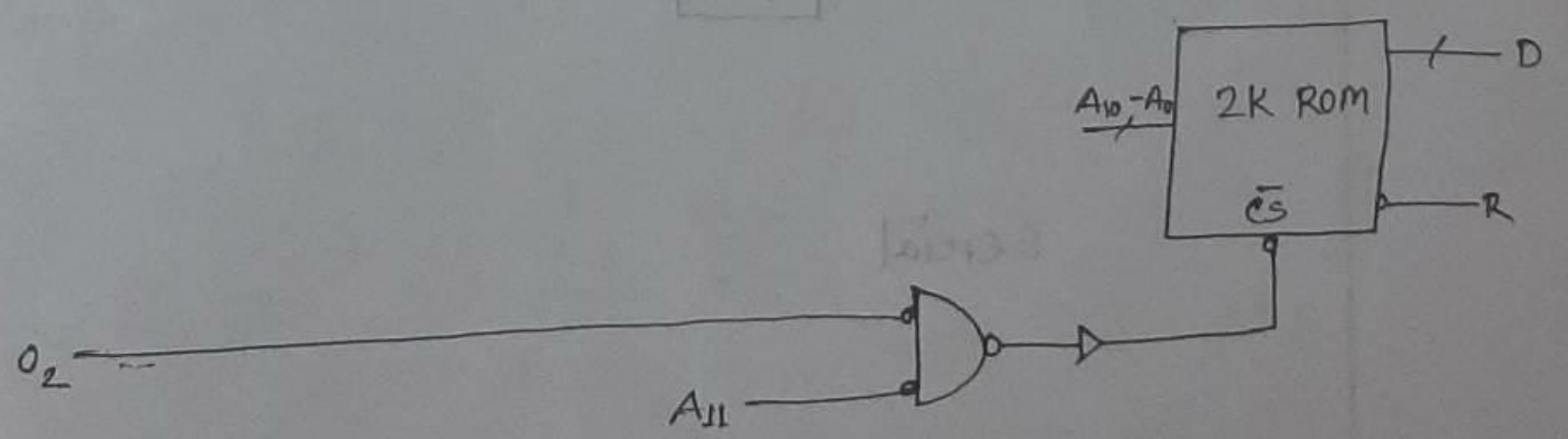
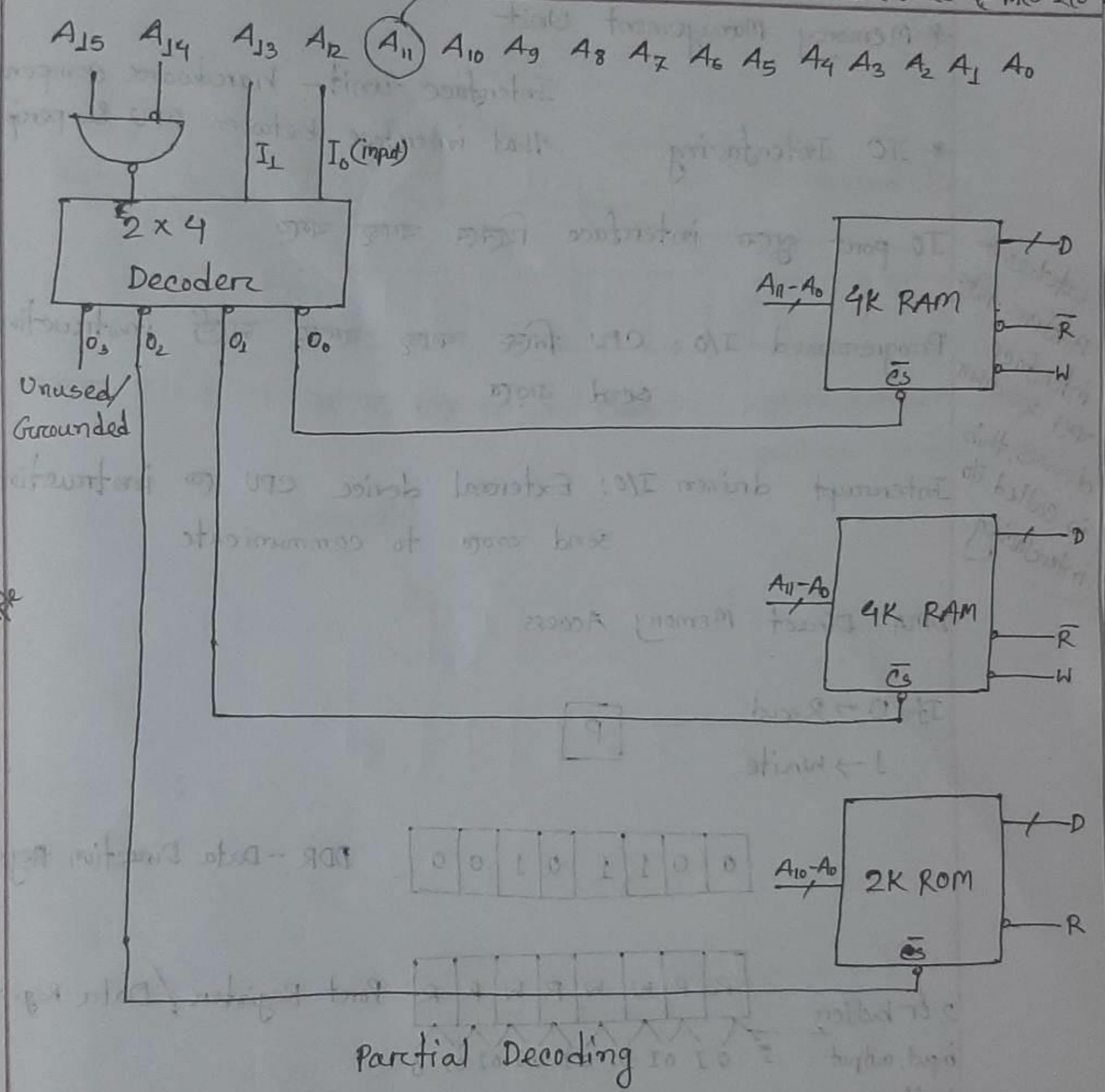
Address Table
 $A_{15} = 1, A_{14} = 0$
 A_{12}, A_{13} are
 value change
 2 bits IP for
 2 bits

Don't care
 address - 3 multiple

2K

Base 4KB
 $8000H + (2^{10} \times 4 - 1)$
 $= 8000H + (1023)_{10}$
 $= 8000H + 03FFH$
 $= 83FFH$
 $A_2 A_1 A_0$
 0 0 0 = 8000H
 1 1 1 = 8FFFH
 0 0 0 = 9000H
 1 1 1 = 9FFFH
 0 0 0 = A000H-AFF
 1 1 1 = A800H-AFFF

Full decoding जहाँ हम = Unused addresses जहाँ यह सर; value fixed कर मिले 20



Full Decoding

* Memory Management Unit

Interface unit - hardware components that interface between CPU & peripherals

* IO Interfacing

IO port গুলো interface গিয়েছে কাজ করতে

Programmed I/O: CPU নিজে কাজ করতে সরাসরি instruction send করে

Interrupt driven I/O: External device CPU কে instruction send করে to communicate

DMA - Direct Memory Access

If: 0 → Read

1 → Write

P

0	0	1	1	0	1	0	0
---	---	---	---	---	---	---	---

DDR - Data Direction Register

R	R	W	W	R	W	R	R
---	---	---	---	---	---	---	---

Port Register / Data Register

2 bit buffers
input, output
buffers

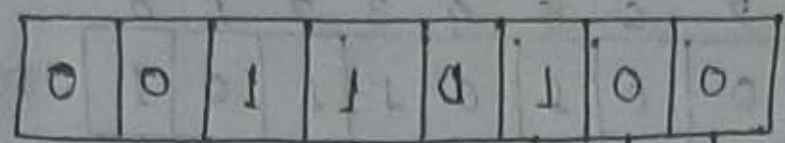
I	0	I	0	I	0	I	0	I	0	I	0
---	---	---	---	---	---	---	---	---	---	---	---

I/O

Serial

P

type = 1
type = 0

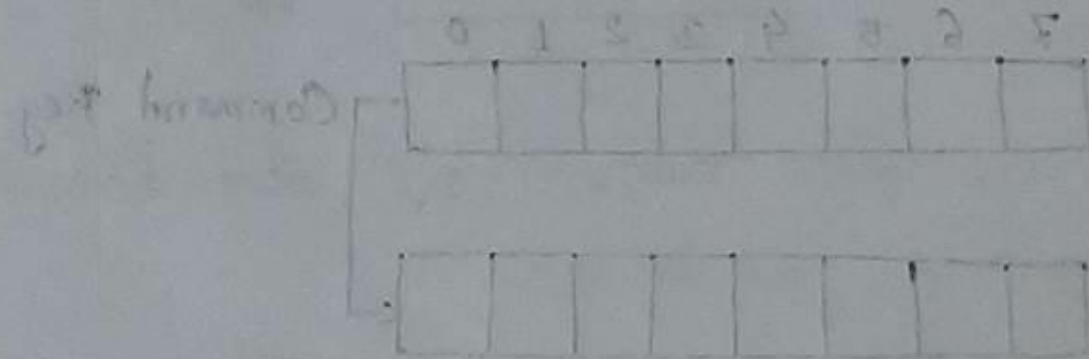


Command Reg



প্রত্যেক 8 bit এর সাথে একেবারে port connected. প্রত্যেক position একেবারে port কে control করে

parallel

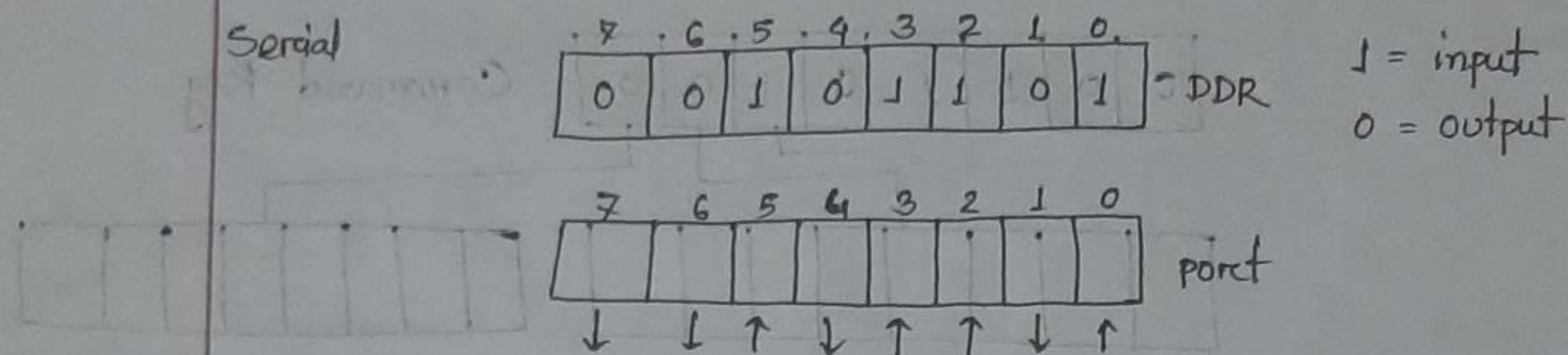


011 Command Reg

011 Command Reg
011 Command Reg
011 Command Reg

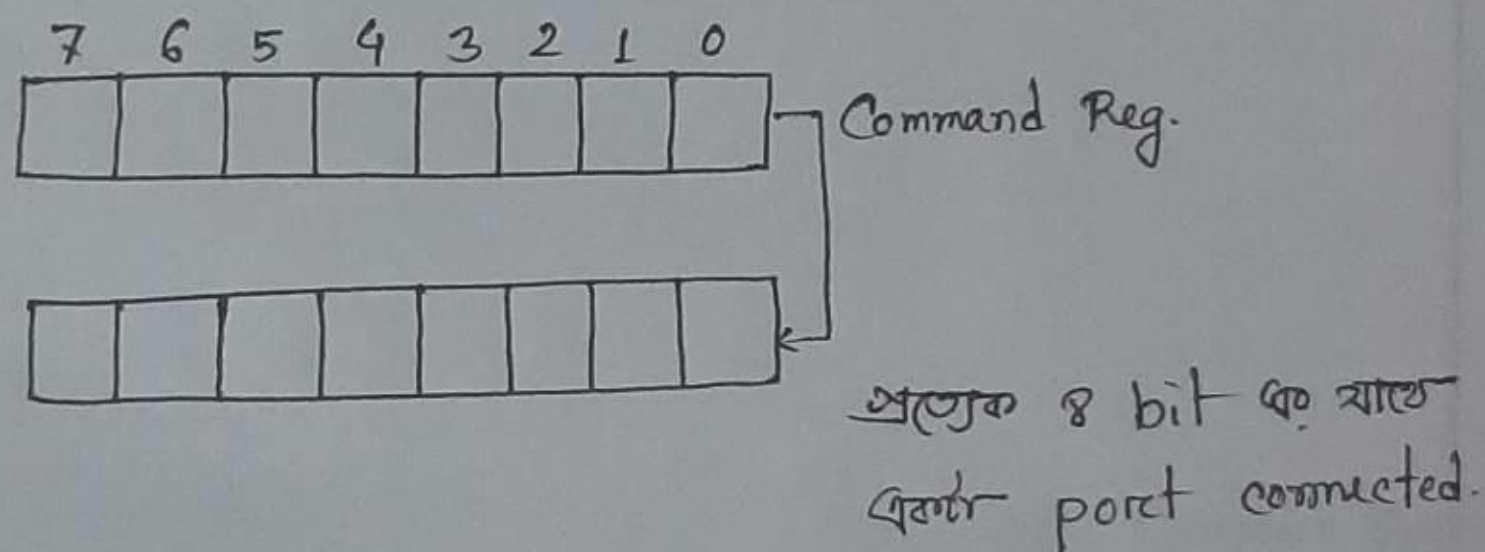
011 Command Reg
011 Command Reg

011 Command Reg
011 Command Reg
011 Command Reg



প্রত্যেক bit ৬০ আলাদা প্রত্যেক bit connected

Parallel



1. Programmed I/O

- i) Isolated I/O - pin দিয়া লোকা যায় m/i/o use হবে
- ii) Standard I/O
- iii) Port

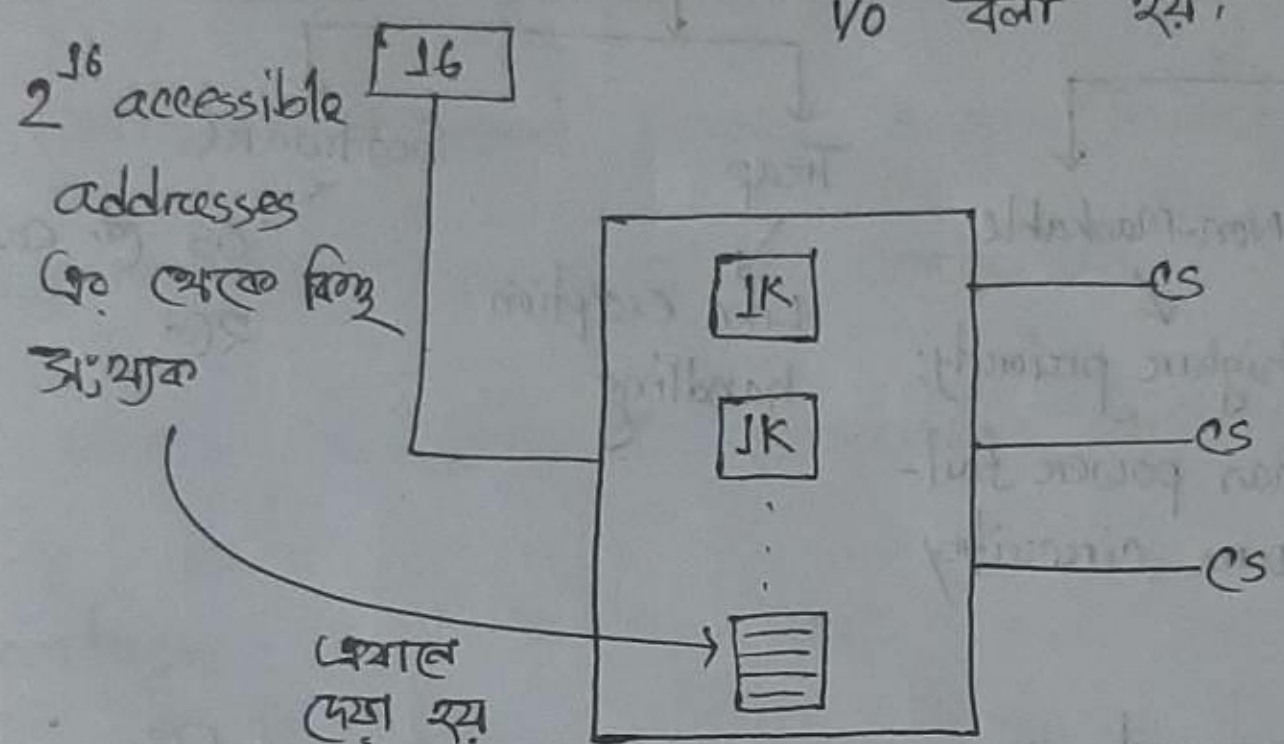
IO/M এর pin ব্যবহার standard;

1 → I/O

0 → M

isolated এর আলাদা specific register থাকবে হয়, প্রকৃষ্ট হাউজ
বান্ধে বান্ধে না - অন্য reg. এর আলাদা বান্ধে বান্ধে না

2. Memory mapped I/O - total accessible address को लेकर बिंदु
 अंशुक i/o को दिए बाकिर memory
 को देया হয় তাই এক memory mapped
 i/o বলা হয়।



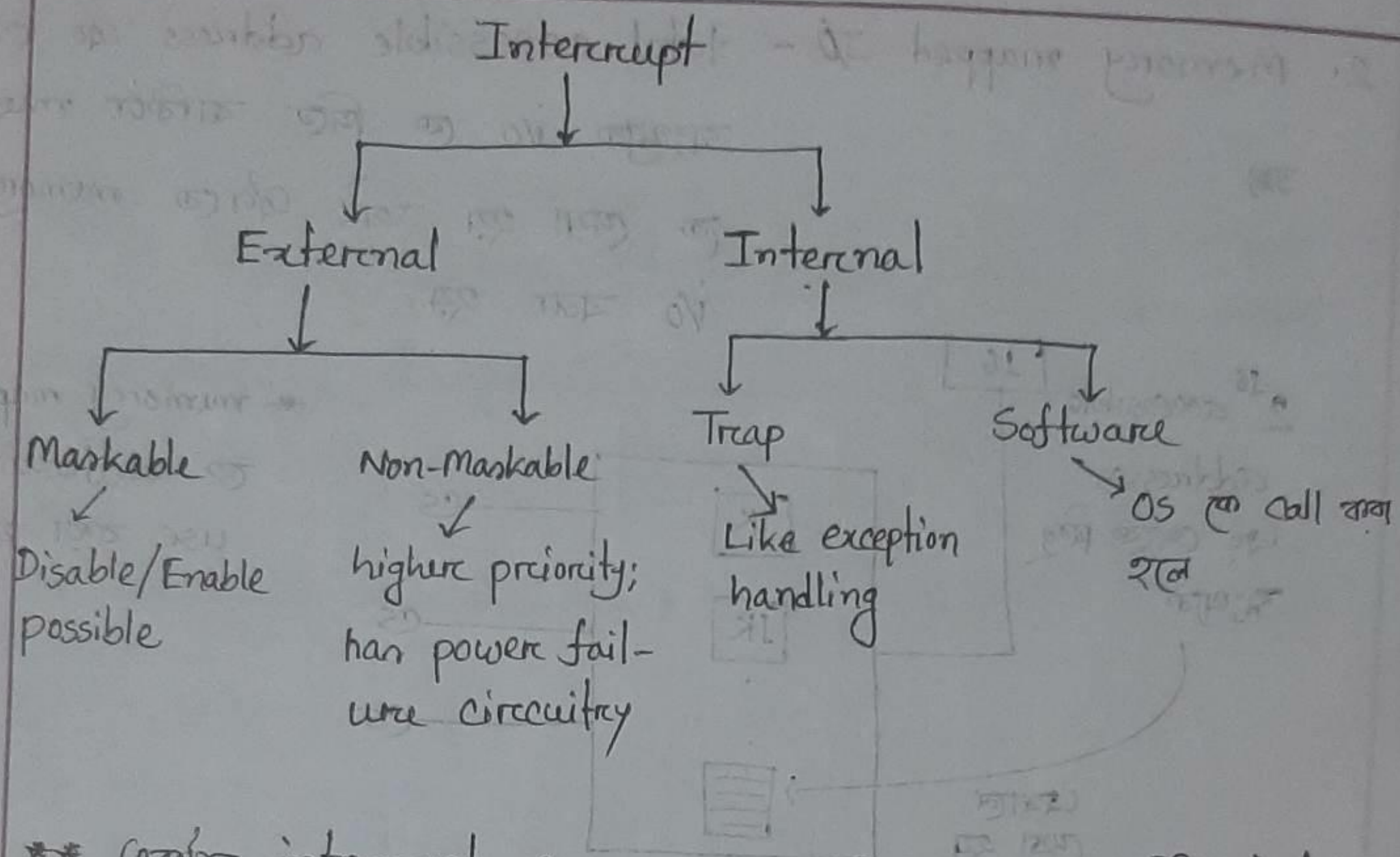
* memory mapped i/o
 to different reg.
 use করা যায়।

i) Unconditional I/O: যেকোন অংক I/O device को পাঠানো থাকে;
 I/O device को always active/ready থাকে
 হয়; power consumption বেশি

ii) Conditional I/O: busy pin থাকে; handshaking को through
 communicate করে; wait করে দেখে I/O
 device free আছে কিনা; processor को বাত বাত
 check করতে হয়।

বাত বাত check করতে যেন বা হয় তাই -

3. Interrupt driven I/O: device initiate করে; int pin থাকে
 লক্ষ্যে on করে communicate করতে হয়



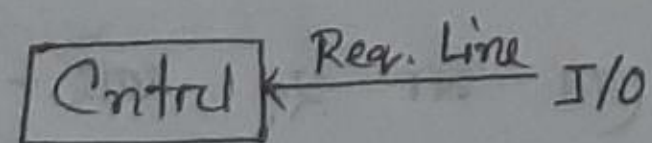
** External interrupt पर जहाँ processor के एकसर्विक int pin प्राप्त होता है, वहाँ priority दिए जा Depend

करते -

- Poll interrupt: depends on processor

- Daisy chain interrupt: यहाँ processor पाठाने के बाद priority-र सुलाने पाठाने

4. DMA - Direct Memory Access



i) Block transfer - blockwise transfer करते

ii) Cycle stealing - जबकि processor then i/o

iii) Interleaved - यहाँ processor bus use करते हैं और device bus use करते हैं।